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Fall 2026 **Course:** COMPSCI 566

Monocular Depth Estimation: Benchmarking and a Defocus-Informed Refinement

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Progress Summary

The full pipeline is implemented and evaluated end to end. Phase 1 is complete: DA-V2 ViT-S is benchmarked on all 654 NYU-Depth v2 test images, and the classical depth-from-focus baseline (modified Laplacian focus measure, parabolic sub-frame interpolation, confidence map) runs on rendered focal stacks and all 120 DDFF test stacks. Phase 2 is implemented and in ablation: the robust scale fit and the regularized local correction are working on the synthetic scenes, and the combined system runs qualitatively on real DDFF stacks.

Current Results

NYU-Depth v2 test split (654 images), DA-V2 ViT-S, two alignment variants against ground truth:

Alignment	AbsRel	RMSE (m)	δ_1
Least-squares, scale+shift	0.095	0.431	0.912
Median ratio, scale only	0.507	2.376	0.376

The affine fit matters: dropping the shift term roughly doubles AbsRel, which supports modeling $q = s\hat{d} + t$ rather than rescaling alone.
Synthetic focal stacks (10 frames, focus 0.9–7 m, thin-lens blur model, means over four scenes), metric depth scored without alignment:

Method	AbsRel	RMSE (m)	δ_1
Focus baseline (DFF)	0.202	1.521	0.706
DA-V2 aligned (oracle)	0.255	1.296	0.567
DA-V2 + scale fit from focus	0.284	1.508	0.502
Full refinement	0.248	1.530	0.560

The focus baseline is accurate where texture exists (δ_1 0.96–1.00 on textured regions) but degrades on flat regions, exactly where the confidence map reads low ($\bar{w} \approx 0.06$–0.25 textureless vs. 0.26–0.66 textured). On the calibrated scenes' textured pixels the refinement improves on the focus baseline (AbsRel 0.154 vs. 0.158) and beats the affine-only fit overall (0.248 vs. 0.284), i.e. the local correction r contributes beyond the global scale terms. DA-V2 is weaker than the focus baseline on our synthetic scenes even with oracle alignment — plausible, since the renderer is far from its training distribution — which makes this benchmark a useful stress test rather than a real-data statement.

Difficulties

- **DDFF labels.** The public test archive ships 120 focal stacks but no ground truth (withheld for the benchmark server), and per-frame focus distances are not recorded. We reconstructed focus distances from the refocus geometry in the paper (disparities linear in $[0.02, 0.28]$ px,

$z \approx 0.14/d$) and limited DDFF to qualitative evaluation; measured confidences on the real Lytro stacks are low ($\bar{w} \approx 0.03$–0.05), and scale calibration was rejected on half of the stacks we tried.

- **NYU test split sourcing.** Both canonical sources were unreachable (the DenseDepth mirror returns 403 and the NYU host refuses connections); we used a mirror of the same 654-image Eigen test split and verified image count, units, and depth range.
- **Defocus veiling.** In the layered renderer, blurred near-field texture bleeds over flat surfaces, which lifts the focus measure there (the synthetic corridor's flat ceiling reaches $w \approx 0.4$). This is a genuine failure mode of the focus measurement rather than a renderer artifact and it caps how far we can trust w at occlusion boundaries.
- **Rejected fits.** On scenes where the learned prediction anticorrelates with the focus estimate, the scale fit is degenerate ($s \leq 0$); we fall back to the bounded uncalibrated output per the proposal rather than emit bad metric depth.

Changes from Proposal

DDFF 12-Scene moved from a quantitative to a qualitative dataset: the verification the proposal made conditional showed the released test stacks carry no labels. Rendered scenes now carry the quantitative refinement evaluation, which also gives us controlled ablations. The method itself is unchanged; we added a degenerate-fit guard so uncalibrated output is reported as relative depth.

Next Steps

- First-pass ablations are in (stack size, confidence weighting, λ, μ); next is a tuning/held-out scene split so the reported numbers are not tuned on the evaluation scenes.
- Extend the DDFF qualitative evaluation and, time permitting, phone-captured focal stacks as a second real source.
- Finish the results figures and tables for the project webpage; keep the presentation deck in step with the final numbers.